

	Course	Databases and Information Systems			2026
	Exercise Sheet	2			
	Points	–			
	Release Date	June 2nd 2026	Due Date	June 8th 2026	

Exercise Sheet 2: NoSQL Databases

With this exercise sheet, you prepare a part of the demo to be presented at the end of the semester. This requires working practically. Unfortunately, our exercise rooms this year are not equipped with PCs and the PC pools are not entirely usable, yet, so please bring a laptop or tablet if you want to work in the exercise rooms or need to show something to explain a question. If you do not have any portable device you can use for the exercise, please consider using the lending service of the General Students' Committee¹.

Please read the whole exercise sheet before starting to work. Take notes after every executed step and save your source code for being able to answer the questionnaire in section 3. All questions must be answered reasonably to pass the exercise. However, the answers are not graded as they are highly individual to the setup you are using. Please continue working in the same groups as in the first exercise sheet. You are free to use a programming language of your choice, but be aware that there might be no support if you chose an exotic language.

1 Decide for and set up a NoSQL database

In this part of the exercise, you will download a dataset containing recipes for coffee specialties. These recipes correspond to the coffee sales in the dataset from exercise sheet 1.

- Download the dataset 'coffee recipes'² from moodle
- Open the file in a text editor and get an overview of the contained information
- Install a **NoSQL** database which is capable of querying the data in the file
- Use the API of your database** to find the ingredients for a recipe called "Americano with Milk"

2 Connection with Relational Data

In this part of the exercise, the new data is connected with the data from exercise sheet 1.

- The new dataset is not complete. Find out which coffee specialty from the sales table is missing in the recipe data.
- Insert an entry for the missing data into the recipe data set. Realism is nice but not required.
- Once there is a recipe for each sold coffee, compute the aggregated time per day that was used for coffee preparation. Document the individual steps for solving this task.

¹<https://www.asta-uhh.de/4-service/1-verleihservice.html>, accessed 15/04/2026

²<https://lernen.min.uni-hamburg.de/mod/resource/view.php?id=282028>

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3 Questionnaire

Go to moodle and answer the questionnaire 'Questionnaire for Exercise Sheet 2'³. **It is sufficient if one member of a group answers the questionnaire, even if moodle does not mark the task as completed for the other group member(s).**

³<https://lernen.min.uni-hamburg.de/mod/feedback/view.php?id=282034>